

CareerCode Academy

JavaScript Algorithms & Data Structures

Course Overview

Master JavaScript fundamentals, ES6, debugging, OOP, algorithms, and data structures.

Module 1: JavaScript Fundamentals

Variables, data types, operators, conditionals, loops, functions, scope, and arrays.

Module 2: ES6 and Modern JavaScript

Arrow functions, destructuring, spread operator, modules, promises, and async/await.

Module 3: Debugging & Problem Solving

Console debugging, error handling, algorithmic thinking, and code optimization.

Module 4: Object-Oriented Programming

Objects, classes, inheritance, encapsulation, and prototypes.

Module 5: Data Structures

Arrays, stacks, queues, linked lists, trees, hash tables, and graphs.

Module 6: Algorithms

Searching, sorting, recursion, dynamic programming, greedy algorithms, and graph traversal.

Module 7: Interview Preparation

Coding challenges, whiteboard problems, and technical interview strategies.

Projects

Palindrome checker, Roman numeral converter, cash register, sorting visualizer, and path-finding project.

Assessment

Theory test, coding assignments, algorithm challenges, and final project.

Suggested 8-Week Schedule

Week 1: JavaScript Basics

Week 2: Functions and Arrays

Week 3: ES6 Features

Week 4: Object-Oriented Programming

Week 5: Core Data Structures

Week 6: Sorting and Searching Algorithms

Week 7: Advanced Algorithm Challenges

Week 8: Final Project and Assessment